INTERNSHIP REPORT

ON

PIR MOTION SENSOR SECURITY SYSTEM USING MICROCONTROLLER

(TASK-2)

IN THE DOMAIN OF

EMBEDDED SYSTEMS AND IOT

BY

MANOGYNA.R

AT

MAINCRAFTS TECHNOLOGY

2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Main Crafts Technology for providing me the opportunity to work on this internship task. I am thankful to my mentors and supervisors for their guidance, support, and valuable suggestions throughout the completion of this project. I also extend my thanks to my faculty members and friends who helped me during the development and simulation of this project using Wokwi platform.

ABSTRACT

This project demonstrates a PIR (Passive Infrared) Motion Sensor-based Security System implemented using a microcontroller in the Wokwi simulation environment. The system detects human motion by sensing changes in infrared radiation and triggers an alert using LED and buzzer. The main objective is to design a low-cost and efficient security system using embedded systems concepts. The simulation verifies the working of motion detection and alert mechanism, making it suitable for real-time security applications such as homes, offices, and industries.

INDEX

1. INTRODUCTION

A PIR Motion Sensor Security System is an embedded system used to detect human movement and provide security alerts. Security systems are widely used in homes, offices, and industries to improve safety and prevent unauthorized access. PIR sensors are commonly used because they can detect motion by sensing infrared radiation emitted from the human body.

In this project, Arduino UNO is used as the microcontroller to control the entire system. The PIR sensor detects motion and sends signals to the Arduino UNO. An LED is connected as an output indicator, and a resistor is used to limit the current flow and protect the LED from damage. These components work together to create a simple motion detection system.

The project was designed and tested using the Wokwi simulation platform. Whenever motion is detected by the PIR sensor, the Arduino UNO processes the signal and turns on the LED to indicate detection. This project helps in understanding sensor interfacing, microcontroller programming, and the basic working of embedded security systems.

2. OBJECTIVES

- To design a motion detection security system using a PIR sensor.
- To interface a PIR sensor with a microcontroller.
- To detect human motion using infrared sensing technology.
- To activate LED and buzzer when motion is detected.
- To simulate the complete system using Wokwi simulator.

3. COMPONENT USED

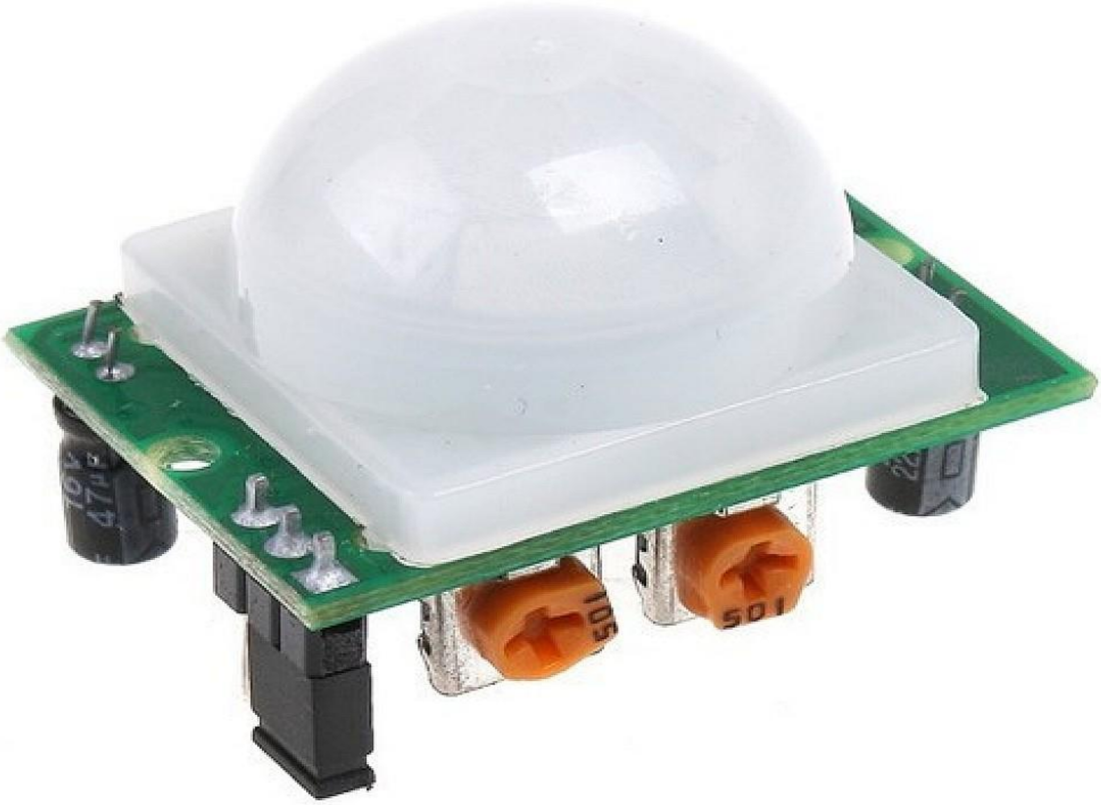
A PIR Motion Sensor Security System is used to detect human movement and provide security alerts. In this project, Arduino UNO is used as the microcontroller to control the system operations. The PIR sensor detects motion by sensing infrared radiation changes from the human body. An LED is used as an indicator to show motion detection, and a resistor is connected to protect the LED by limiting the current flow. The project was simulated using the Wokwi platform to understand the working of embedded systems and motion detection applications. The following Components are:

(i) Arduino

Arduino UNO is the main microcontroller used in this project to control the complete motion detection system. It receives input signals from the PIR motion sensor and processes them according to the programmed instructions. Based on the sensor output, the Arduino UNO activates the LED indicator when motion is detected. It is built on the ATmega328P microcontroller and is widely used in embedded systems and automation projects due to its simplicity and ease of programming.

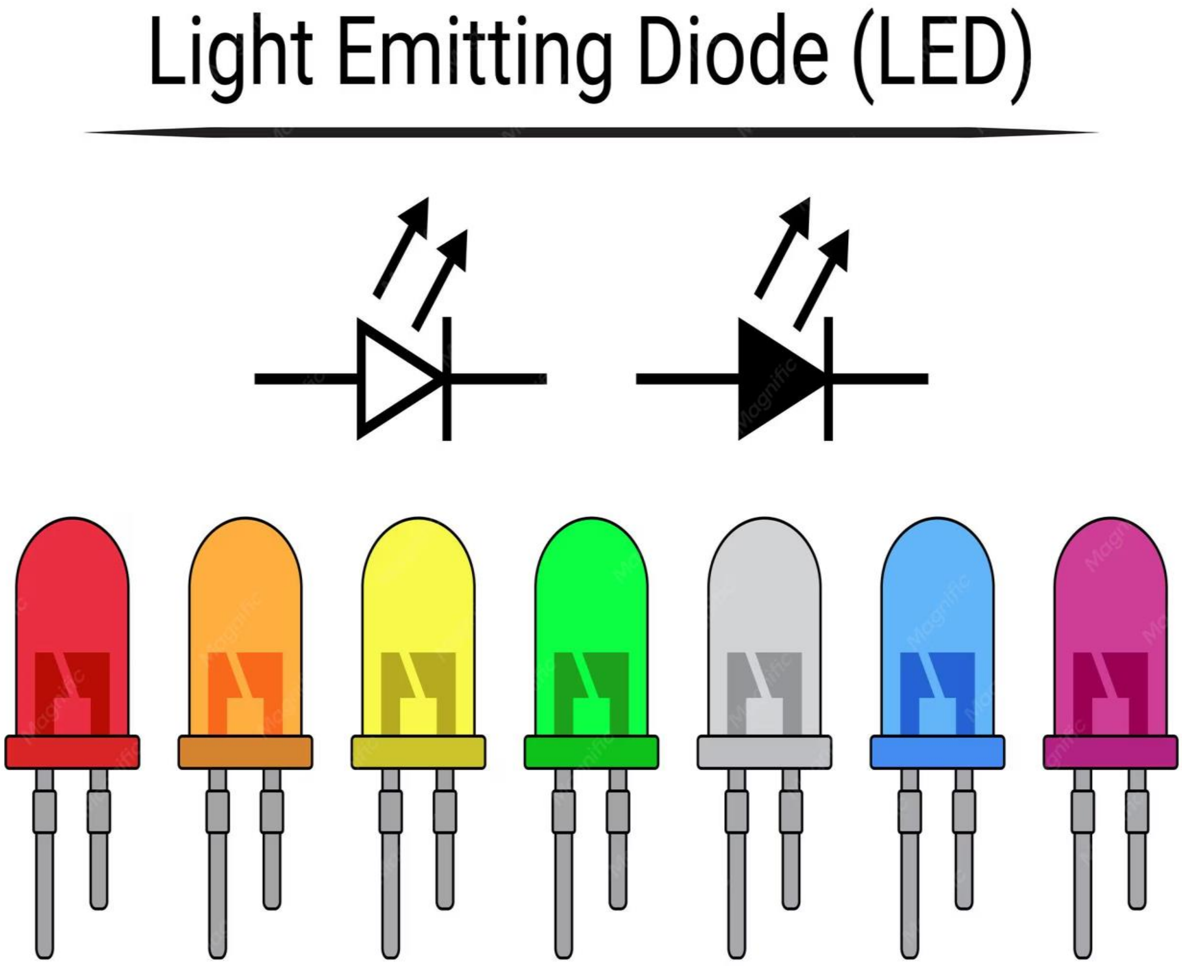
(ii) PIR Motion Sensor

The PIR (Passive Infrared) Motion Sensor is used to detect human movement in the surrounding area. It works by sensing changes in infrared radiation emitted by living bodies such as humans. Whenever a person moves within the detection range, the sensor generates a signal and sends it to the Arduino UNO. PIR sensors are commonly used in security systems because they consume low power and provide reliable motion detection. In this project, the PIR sensor acts as the main sensing device for detecting intrusion or movement.



(iii) LED

The LED (Light Emitting Diode) is used as an output indicator in this project. It glows whenever the PIR sensor detects motion and sends a signal to the Arduino UNO. LEDs are commonly used in electronic circuits because they consume less power and respond quickly. In this project, the LED acts as a simple alert device for security indication.



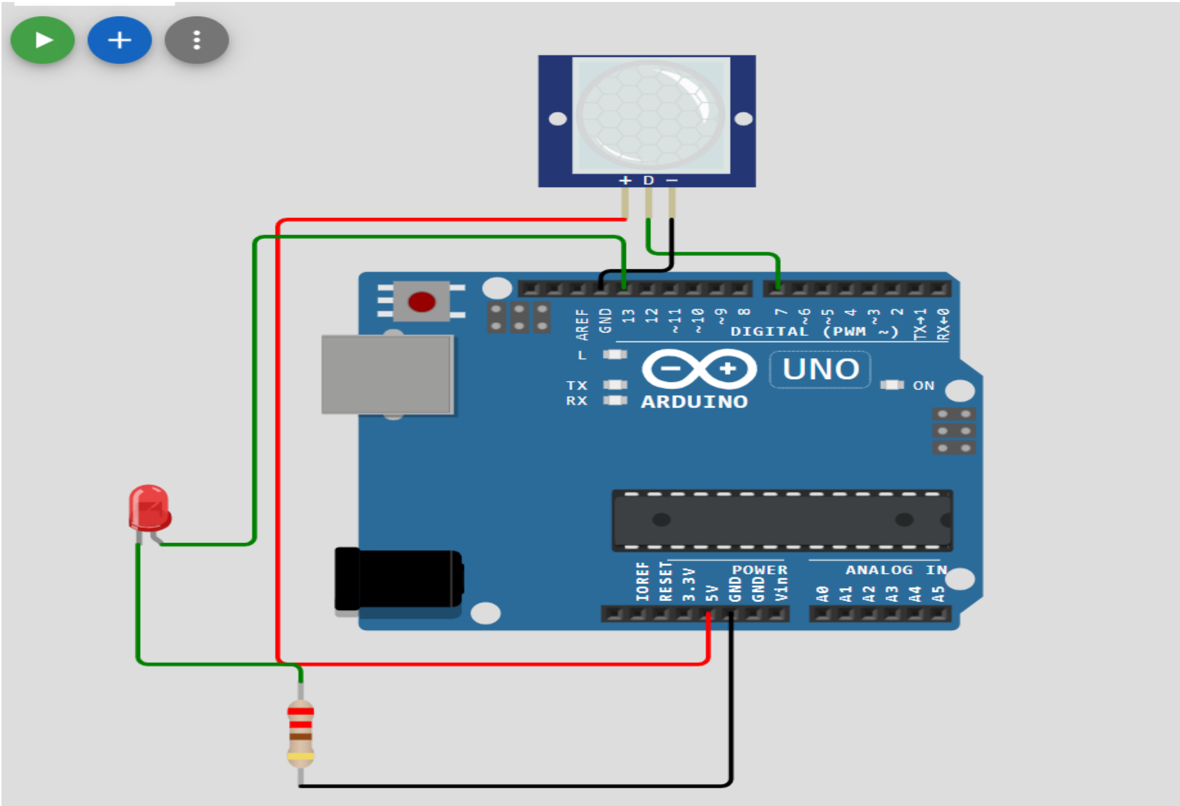
(iv) Resistor

A resistor is an electronic component used to control and limit the flow of electric current in a circuit. In this project, the resistor relates to the LED to protect it from excess current that may damage the component. It helps in maintaining safe current levels and ensures proper operation of the LED. The resistor used in this project improves the reliability and safety of the circuit.

4. Working Principle

1. The PIR motion sensor continuously monitors the surrounding area for infrared radiation changes.
2. Human bodies emit infrared radiation, and when a person moves near the sensor, the infrared levels change.
3. The PIR sensor detects this change and generates an output signal.
4. The output signal from the PIR sensor is sent to the Arduino UNO microcontroller.
5. The Arduino UNO processes the received signal according to the programmed instructions.
6. When motion is detected, the Arduino UNO turns ON the LED indicator.
7. The resistor connected with the LED limits the current flow and protects the LED from damage.
8. The entire system was simulated and tested using the Wokwi simulation platform.

5. CIRCUIT DIAGRAM



The circuit diagram of the PIR Motion Sensor Security System consists of an Arduino UNO, PIR motion sensor, LED, and resistor. The VCC pin of the PIR sensor is connected to the 5V pin of the Arduino UNO, and the GND pin is connected to the ground terminal. The OUT pin of the PIR sensor is connected to one of the digital input pins of the Arduino UNO to send motion detection signals.

The LED is connected to a digital output pin of the Arduino UNO through a resistor. The resistor is used to limit the current flow and protect the LED from damage. When the PIR sensor detects motion, the Arduino UNO processes the signal and turns ON the LED indicator. The complete circuit was designed and simulated using the Wokwi simulation platform.

6. ALGORITHM/FLOW

1. Start the system.
2. Supply power to the Arduino UNO and PIR motion sensor.
3. Initialize the PIR sensor pin as INPUT.
4. Initialize the LED pin as OUTPUT.
5. The PIR sensor continuously monitors the surrounding area.
6. Detect changes in infrared radiation caused by human movement.
7. Send the detected signal from the PIR sensor to the Arduino UNO.
8. The Arduino UNO processes the received signal.
9. If motion is detected, turn ON the LED indicator; otherwise keep the LED OFF.
10. Repeat the process continuously until the system is stopped.

7. CODE

WOKWI SAVE SHARE

sketch.ino • diagram.json • Library Manager

```
int pirPin = 7;
int ledPin= 13;
void setup()
{
  pinMode(ledPin, OUTPUT);
  pinMode(pirPin, INPUT);
  Serial.begin(9600);
}
void loop()
{
  int motion = digitalRead(pirPin);
  Serial.println(motion);
  if (motion==HIGH)
  {
    digitalWrite(ledPin, HIGH);
  }
  else
  {
    digitalWrite(ledPin,LOW);
  }
  delay(1000);
}
```

CODE EXPLANATION:

1. int pirPin = 7;

   This line declares an integer variable named pirPin and assigns pin number 7 to it. The PIR motion sensor output pin is connected to digital pin 7 of the Arduino UNO.

2. int ledPin = 13;

   This line declares another integer variable named `ledPin` and assigns pin number 13 to it. The LED is connected to digital pin 13 of the Arduino UNO.

3. void setup ()

The setup () function is used to initialize the system settings. It runs only once when the Arduino is powered ON or reset.

4. pinMode(ledPin, OUTPUT);

This line sets the LED pin as an OUTPUT pin because the LED is used to display output indication.

5. pinMode(pirPin, INPUT);

This line sets the PIR sensor pin as an INPUT pin because the Arduino receives signals from the PIR sensor.

6. Serial.begin(9600);

This line starts serial communication at a baud rate of 9600 bits per second. It is used to display sensor readings in the Serial Monitor.

7. void loop ()

The loop () function contains the main program logic. It runs continuously as long as the Arduino is powered ON.

8. int motion = digitalRead(pirPin);

This line reads the signal from the PIR sensor pin and stores the value in the variable named motion.

9. Serial.println(motion);

This line prints the value of the motion variable in the Serial Monitor for monitoring and debugging purposes.

10. if (motion==HIGH)

This condition checks whether motion is detected. If the PIR sensor output is HIGH, it means movement has been detected.

11. digitalWrite (ledPin, HIGH);

This line turns ON the LED when motion is detected.

12. Else

The else block executes when no motion is detected.

13. digitalWrite (ledPin, LOW);

This line turns OFF the LED when no motion is detected.

14. Delay (1000);

This line creates a delay of 1000 milliseconds (1 second) before repeating the loop again.

8. APPLICATIONS

- Home security systems for detecting unauthorized entry.
- Office security and intrusion detection systems.
- Automatic lighting systems that turn ON when motion is detected.
- Smart surveillance and monitoring applications.
- Industrial safety systems for restricted area monitoring.
- Motion detection systems in smart homes and automation projects.
- Energy-saving systems by controlling devices based on human presence.
- Security alert systems in banks, museums, and public places.

9. RESULT

The PIR Motion Sensor Security System was successfully designed and simulated using the Wokwi platform. The PIR sensor was able to detect human motion correctly, and the Arduino UNO processed the signal efficiently. Whenever motion was detected, the LED turned ON as an alert indication. The project verified the successful working of motion detection and embedded security system concepts.

Click here to view simulation video

10. CONCLUSION

The PIR Motion Sensor Security System was successfully designed and simulated using the Wokwi platform. The project demonstrated the working of motion detection using a PIR sensor and Arduino UNO microcontroller. Whenever motion was detected, the LED turned ON successfully, indicating proper system operation. This project helped in understanding embedded systems, sensor interfacing, and basic security applications using microcontrollers.